Graphical Evaluation of Continuous-Time Convolution

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Key Concepts

1) Understanding how to perform convolution graphically is useful for developing insight into system behavior.

2) Continuous-time convolution cannot be evaluated exactly with a computer.

3) The graphical approach proceeds in three steps

$$y(t) = \int_{-\infty}^{\infty} x(t-\tau)h(\tau)d\tau$$

a) Form $x(t-\tau)$ by treating t as a constant and τ as the independent variable.

   i) Graph $x(\tau)$

   ii) Reflect $x(\tau)$ about $\tau = 0$ to obtain $x(-\tau)$.

   iii) Shift $x(-\tau)$ by t to obtain $x(t-\tau)$. This shift puts the value that was originally at $\tau = 0$ at $\tau = t$. For $t < 0$ shift $x(-\tau)$ to the left and for $t > 0$ shift $x(-\tau)$ to the right.

b) Form $x(t-\tau)h(\tau)$ as a function of the independent variable τ for a given value of t.

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c) Integrate $x(t-\tau)h(\tau)$ (over all τ) to obtain $y(t)$.

4) Repeat the three-step procedure for each value of t that is of interest.

5) Common types of intuition that can be obtained with the graphical approach:

   a) When is the output of the system zero?

   b) Does the output increase, decrease, or oscillate?

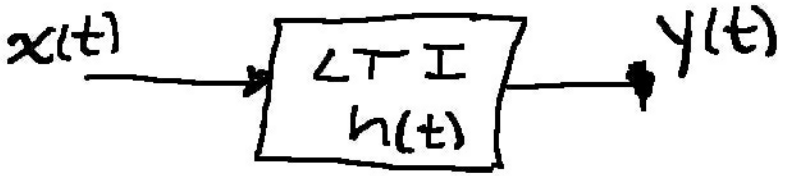
   c) Does the output approach a steady-state?

6) Convolution with impulses is particularly simple due to the sifting property.

$$x(t) * \delta(t - t_0) = x(t - t_0)$$

The signal is shifted by the delay (t_0) of the impulse.

Relates the output of a LTI system to the input using the impulse response



$$y(t) = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau$$

"convolution integral"

Why learn graphical evaluation?

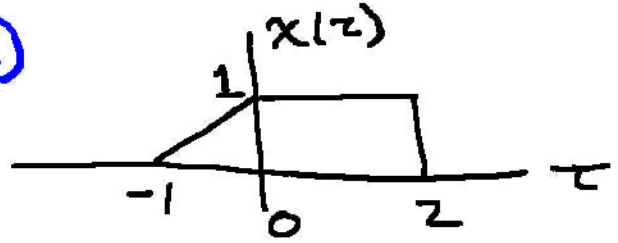
1) Develop insight that allows you to predict/analyze system behavior
2) Can't be evaluated (exactly) with a computer.
3) Convolution occurs in other contexts

$$y(t) = \int_{-\infty}^{\infty} \underbrace{x(t-\tau)}_{1}\, \underbrace{h(\tau)}_{2}\, d\tau$$

3

1) Form $x(t-\tau)$ as a function of τ with t a constant

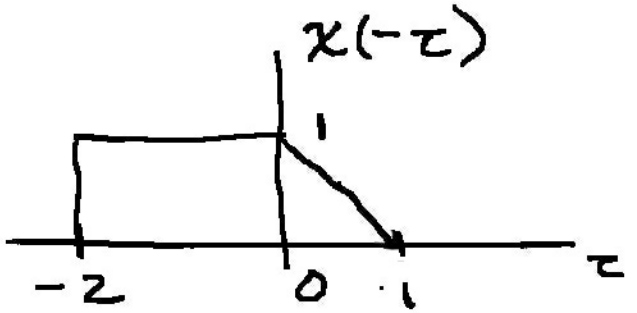
a)



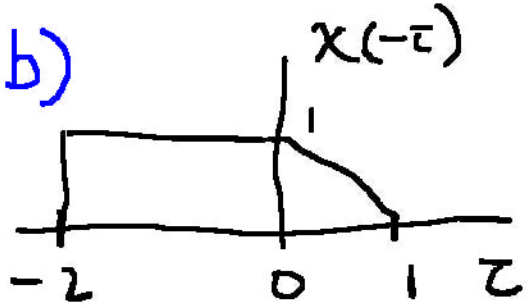
replace τ by $-\tau$

$x(\tau) \rightarrow x(-\tau)$

reflect about $\tau = 0$



b)

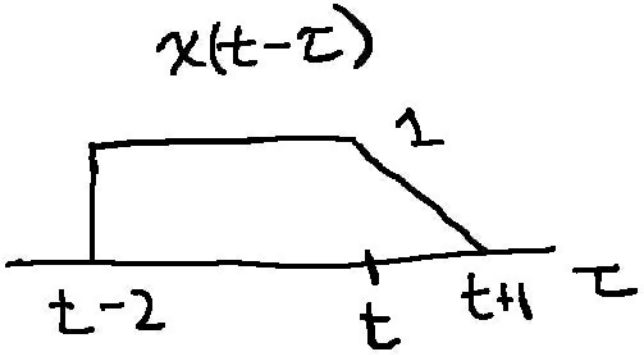


replace τ by $\tau - t$

$x(-\tau) \rightarrow x(-(\tau - t))$

$= x(t-\tau)$

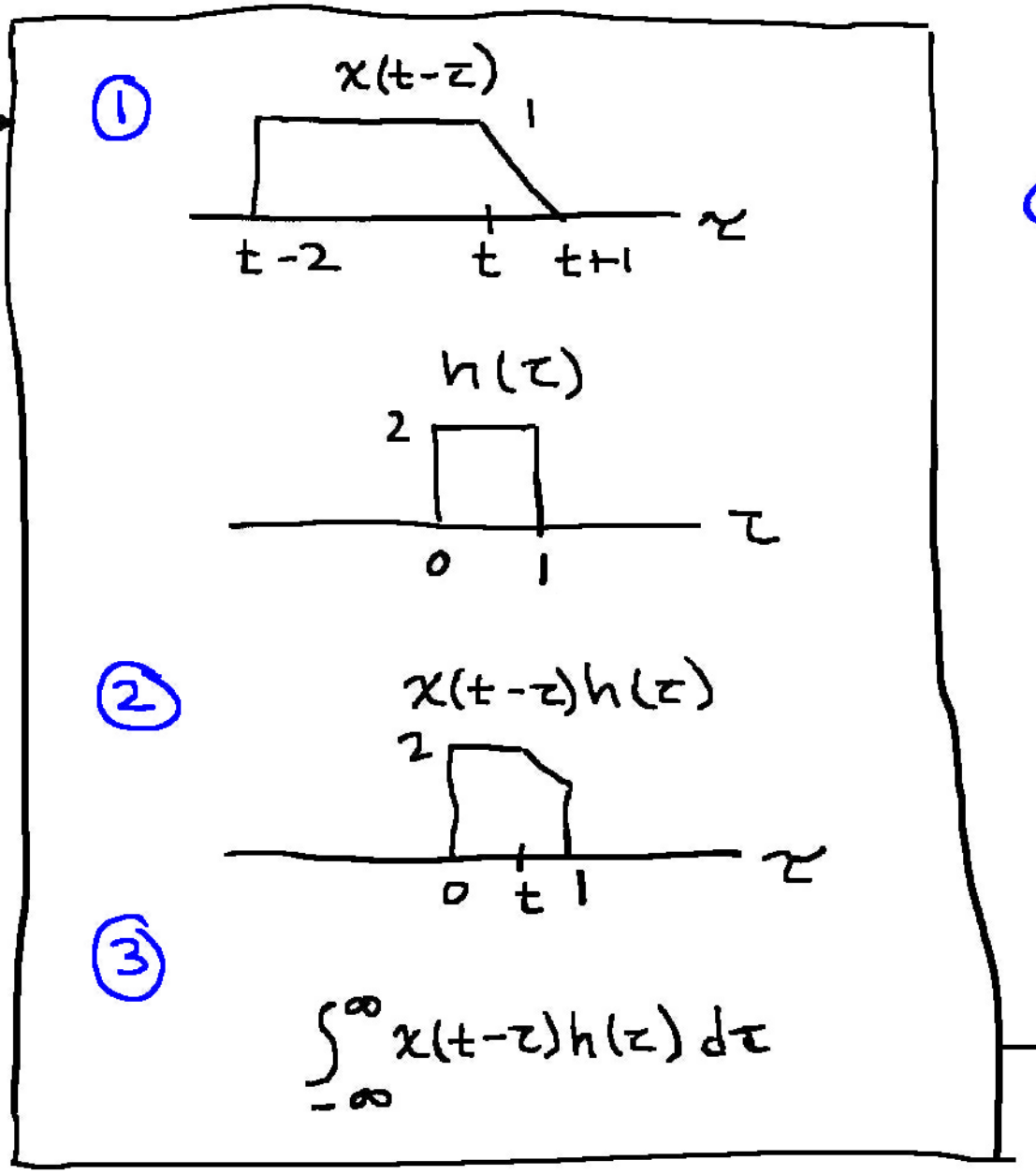
shift $x(-\tau)$ right by t



Treat t as a constant!

Convolution Factory

$x(\cdot)$



$y(t)$

t negative

① shift $x(-\tau)$ by t to get $x(t-\tau)$

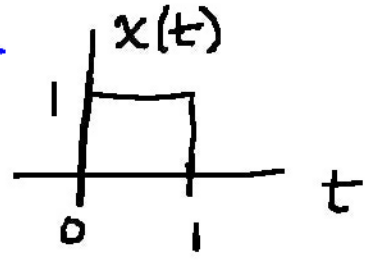
② form $x(t-\tau)\,h(\tau)$

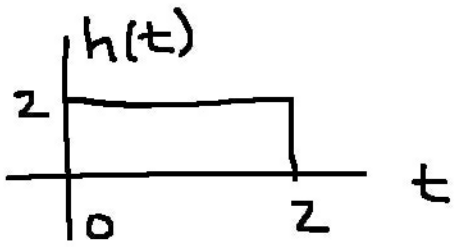
③ find area

$$\int_{-\infty}^{\infty} x(t-\tau)h(\tau)\,d\tau$$

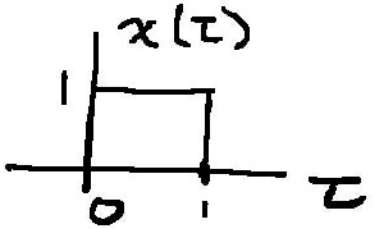
increase t

Example: Find $y(t) = x(t) * h(t)$ for

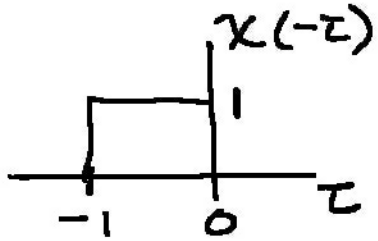




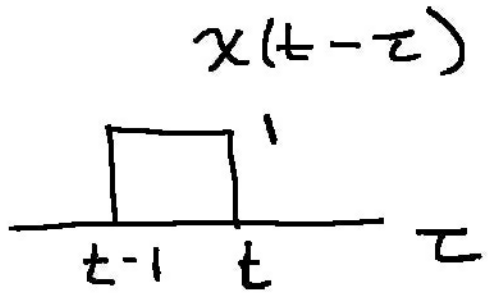
Step 1: Graph $x(t-\tau)$ vs. τ



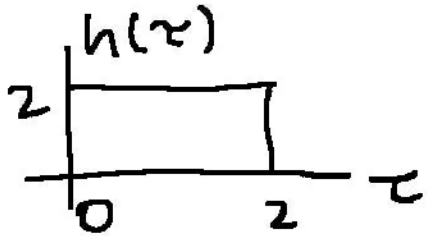
$\tau \rightarrow -\tau$



$\tau \rightarrow \tau - t$



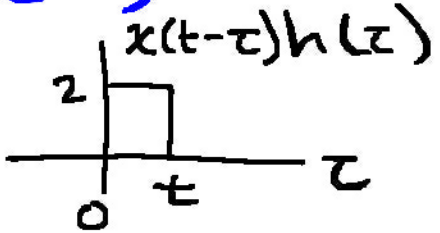
Step 2: Form $x(t-\tau)h(\tau)$



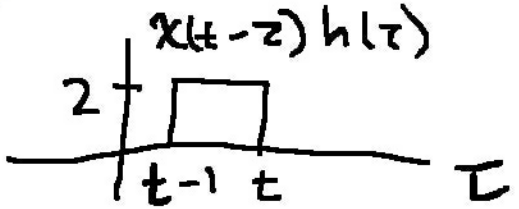
Case a) $t < 0$

$x(t-\tau)h(\tau) = 0$

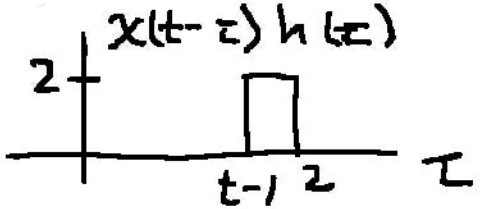
case b) $0 < t < 1$



case c) $1 < t < 2$



case d) $2 < t < 3$



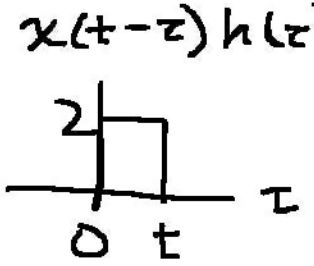
case e) $t > 3$

$x(t-\tau)h(\tau) = 0$

Step 3: Area $y(t) = \int_{-\infty}^{\infty} x(t-\tau)\,h(\tau)\,d\tau$

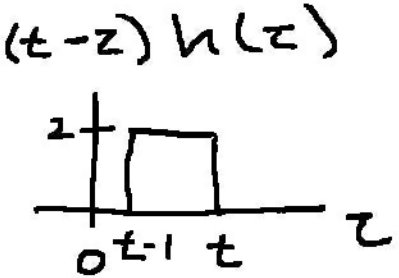
Case a), e) $x(t-\tau)h(\tau)=0 \Rightarrow y(t)=0$ $\quad t<0,\ t>3$

Case b)



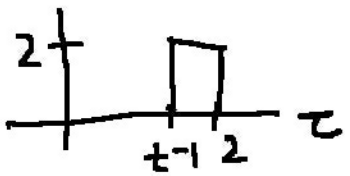
$\Rightarrow y(t)=\int_0^t 2\,d\tau = 2t \quad 0<t<1$

Case c)

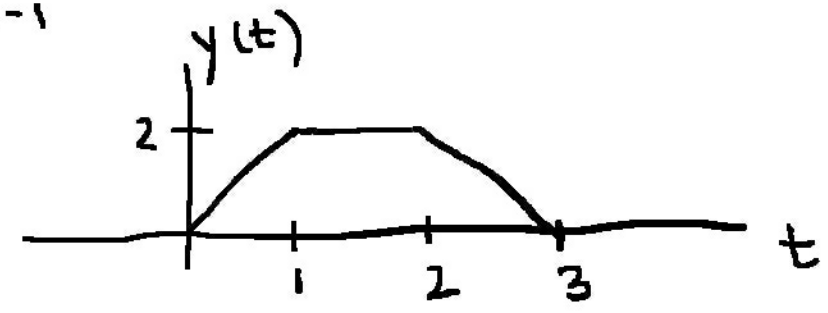


$\Rightarrow y(t)=\int_{t-1}^{t} 2\,d\tau = 2 \quad 1<t<2$

Case d)

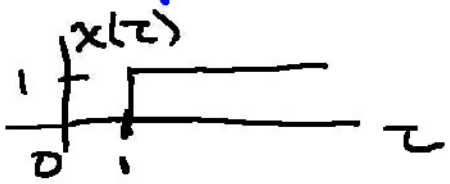


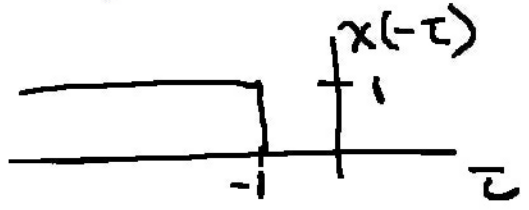
$\Rightarrow y(t)=\int_{t-1}^{2} 2\,d\tau = 4-(2t-2)=6-2t \quad 2<t<3$

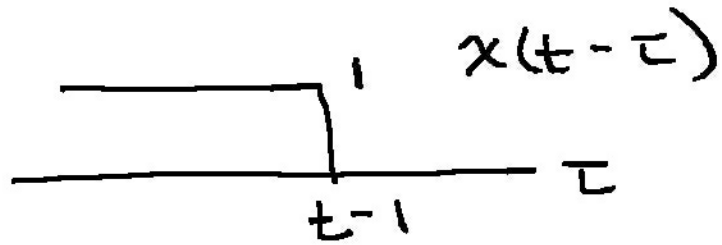


Example: Find $y(t) = x(t) * h(t)$ for $x(t) = u(t-1), h(t) = e^{-t}u(t)$

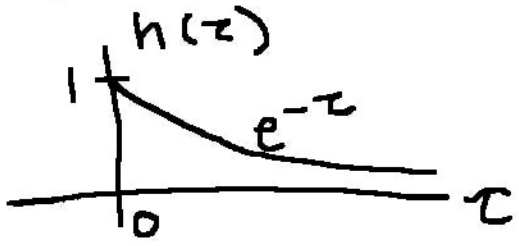
Step 1: Form $x(t-\tau)$ vs. τ





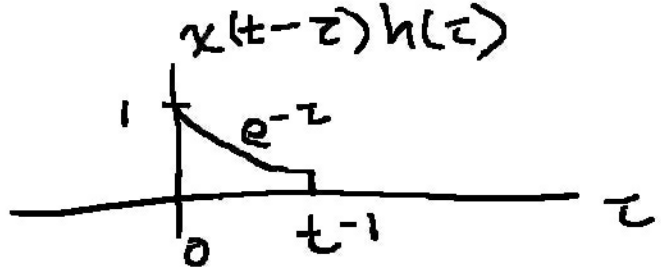


Step 2: Form $x(t-\tau)h(\tau)$



Case a) $t < 1$ $\quad x(t-\tau)\,h(\tau) = 0$

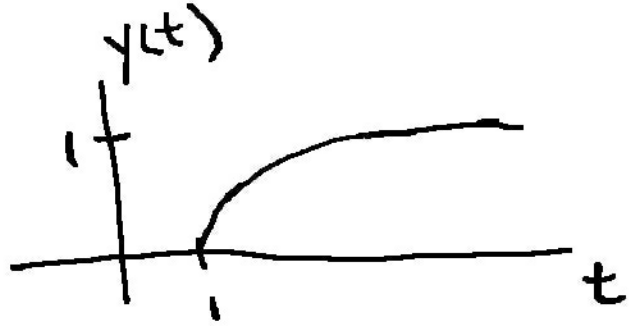
Case b) $t \geq 1$



Step 3: Area $y(t) = \int_{-\infty}^{\infty} x(t-\tau)\,h(\tau)\,d\tau$

Case a) $y(t) = 0, \quad t < 1$

Case b) $y(t) = \int_0^{t-1} e^{-\tau}\,d\tau = -e^{-\tau}\Big|_0^{t-1} = 1 - e^{-(t-1)}, \quad t \geq 1$



Convolution with impulses: simple

$$x(t) * \delta(t - t_0) = x(t - t_0)$$

impulse at t_0: shift signal by t_0

(consequence of impulse sifting property)

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